FIT3080 Artificial Intelligence
Practice Questions in Agents and Problem Solving as Search

1. [**Agents**] Consider the vacuum agent presented in class, but assume that the room has 8 squares and a square can get dirty after it has been cleaned.

   (a) Use PEAS to specify the task environment.

   (b) Specify the attributes of the environment.

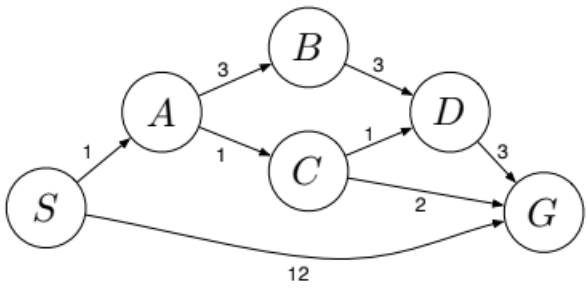
2. [**Agents**] Consider shopping for used AI books on the Internet.

   (a) Use PEAS to specify the task environment

   (b) Specify the attributes of the environment.

3. [**Backtrack**] Specify a state representation, operators and goal test to solve the N-queen problem with N=4:
   Four queens have to be placed on a 4X4 board where no row will have more than one queen, no column will have more than one queen, and no diagonal will have more than one queen.
   Illustrate the workings of the backtracking algorithm (Backtrack1) to solve this problem.

4. [**GraphSearch**]



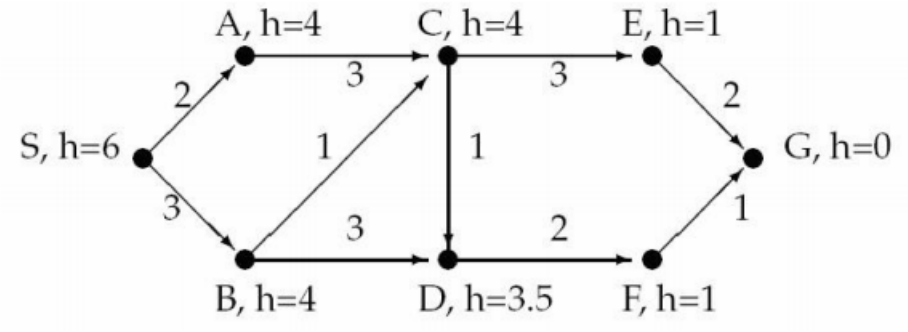
   Answer the following questions about the search problem shown above. For the questions that ask for a path, please give your answers in the form '$S - ... - G$.'

   (a) Breadth-first graph search
   List the nodes according to their order of expansion. For each expansion, list FRONTIER (**with the nodes in the correct order**) and EXPLORED.
   What path would BFS return for this search problem?

   (b) Uniform cost graph search
   List the nodes according to their order of expansion. For each expansion, list FRONTIER (**with the nodes in the correct order**) and EXPLORED.
   What path would uniform-cost return for this search problem?

   (c) Depth-first graph search
   List the nodes according to their order of expansion. For each expansion, list FRONTIER (**with the nodes in the correct order**) and EXPLORED.
   What path would DFS return for this search problem?

5. [**GraphSearch**] Suppose we want to use algorithm A on the graph below to find the shortest path from node S to node G. Each node is labeled by a capital letter and the value of a heuristic function. Each edge is labeled by the cost to traverse that edge.



For this problem:

- Apply algorithm A to this graph, filling in the table below. Indicate the f, g and h values of each node in FRONTIER and EXPLORED as shown in the first two rows of the table.
  If you find a path to a node that is already in FRONTIER, update its cost (using the lower f value) instead of adding another copy of that node to FRONTIER.

iteration	node expanded	FRONTIER	EXPLORED
0		S=0+6=6	
1	S	A=2+4=6; B=3+4=7	S

- Show the path found by algorithm A on the graph above.

6. [**Algorithm A***] The evaluation function $f(n) = d(n) + W(n)$, where $d(n)$ is the cost of arriving at node n and $W(n)$ is the sum of Manhattan Distance for each tile, is used in conjunction with the algorithm A^* to search from the start node:

2	8	3
1	6	4
7		5

To the goal node:

1	2	3
8		4
7	6	5

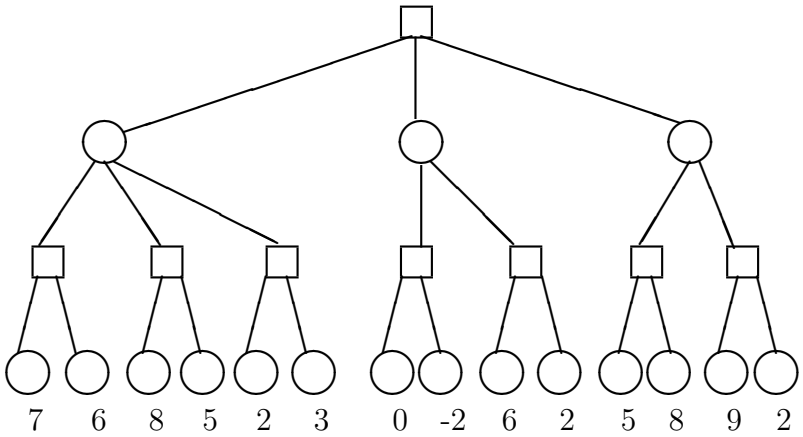
Use this evaluation function to search forward (from start node to goal node) and backward (from the goal node to the start node). Where would the backward search meet the forward search?

7. [**Search**] Which of the following are true and which are false? Explain your answers.

   (a) Depth-first search always expands at least as many nodes as A* search with an admissible heuristic.

   (b) h(n) = 0 is an admissible heuristic for the 8-puzzle.

8. [**Game playing**] Consider the following game tree, where MAX plays in the square positions and MIN plays in the circles.



   (a) Conduct an α-β search of this game starting at the leftmost node to determine which move should MAX make. Draw the resulting game tree so that only the visited nodes appear in your diagram, i.e., **without** the nodes that are cut off. Indicate clearly the backed up value of each node, the updates performed on the backed up values, the α cut-offs and the β cut-offs you have performed.

   (b) What is the best move for MAX and what is its backed up value?

9. [**Game playing**] Consider a game tree with branching factor 2 and depth 5. Assume that it is MAX's turn to play, and that the evaluation function at the leaf nodes is in increasing order from left to right, such that its value for the leftmost node is 1, and for the rightmost node is 16 (the leaf nodes are MAX nodes).

   Conduct an α-β search of this game tree, starting from leftmost-node-first. In your α-β tree, clearly indicate the propagation of the α and β values, the performed cutoffs and the inspected leaf nodes.

   Upon completion of the search, state the final backed-up value of the root node and the recommended move (Left or Right). Also state the number of α and β cutoffs performed, and the number of leaf nodes generated.